

Databases for Big Data

Exercises 7: BigQuery / Dremel

I.Savnik, 13/1/2026.

Exercises

Exercise 1.

Explain how and why Dremel implements disaggregated storage, computation and memory?

Exercise 2.

Describe the main ideas of Dremel architecture and, in particular, the architecture of the query execution system.

Exercise 3.

What is shuffle-based distributed query execution? Explain its function. Which parts of a query are executed in between two shuffle stages?

Exercise 4.

How can Dremel achieve the linear scale-up? What does this mean exactly?

Exercise 5.

Present the approaches to the adaptive query processing in Dremel. Give some examples.